
When do complex-valued neural networks help? A representation-first study of phase, amplitude, and optimization

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Neural Networks in Complex Spaces project

github.com/ashu1069/Neural-Networks-in-Complex-Spaces

Abstract

We show that a widely cited class of complex-vs-real neural-network performance gaps is, in part, a hyperparameter-selection artifact. On a controlled 3-class RadioML 2018.01A subset, a 22.94 pp matched-shared-trial gap under `crelu` collapses by $9.3\times$ to 2.46 pp under independent per-family tuning, on the same data and the same 16-trial search box. Per-step gradient telemetry traces the difference to an explosion-into-dead-region failure of real baselines at the high-learning-rate first AdamW step, a regime the complex family tolerates because its (Re, Im) parameter coupling distributes the loss signal across more readout slots. A 2×2 learning-rate / activation factorial confirms the threshold is hyperparameter-driven, not activation-driven.

The methodological finding is embedded in a representation-first study of when complex-valued networks help at all. A complex sample $z = x + iy = re^{i\theta}$ packages amplitude and phase into one number; a real network can see it as Cartesian, polar, magnitude-only, or phase-only coordinates, and the right question is which mechanism — representation, the $U(1)$ -equivariant subspace constraint of complex multiplication, or optimization tolerance — is doing the work. On synthetic RF, PSK-only data favor the complex model (0.821 vs 0.333 for magnitude-only), QAM-only data favor magnitude, mixed PSK+QAM gives a small complex edge, and unseen carrier-phase rotation breaks coordinate-dependent models unless augmentation is added. Two application pilots support the same reading: in quantum wavefunctions, momentum is invisible to $|\psi|$ but recoverable from phase; in EEG analytic signals, phase locking, amplitude bursts, and phase-amplitude coupling each select different coordinate views. We frame complex activations through the Wirtinger calculus and a Liouville trilemma (bounded, holomorphic, nonconstant — pick two), and characterize the six standard activations against it. The paper’s claim is consequently scoped: complex-valued models are useful structured inductive biases for phase- and amplitude-bearing data, but their advantage is representation-, symmetry-, and optimization-dependent — and apparent CVNN-vs-real gaps should be reported under both selection rules before being interpreted as architectural.

1 First-principles setup

1.1 What a complex input contains

A complex number is a two-dimensional real object with extra structure. The same point can be written in Cartesian form $z = x + iy$ or polar form $z = re^{i\theta}$, where $r = |z| = \sqrt{x^2 + y^2}$ is magnitude and $\theta = \arg z$ is phase. Equal magnitude does not mean equal information: 5 and $3 + 4i$

both have magnitude 5, but their phases differ, their real parts differ, and a model that only sees $|z|$ cannot distinguish them. Conversely, if the label depends only on energy, phase may be a nuisance rather than a signal.

This simple observation is the reason a complex-vs-real comparison can be misleading. A real-valued network on (x, y) sees the full Cartesian information. A real network on $(r, \cos \theta, \sin \theta)$ sees magnitude and phase explicitly. A magnitude-only or phase-only network sees a controlled information bottleneck. We use all of these views because they answer different questions: whether information is in amplitude, in phase, in their relation, or in the algebraic prior of complex multiplication.

1.2 What a complex layer constrains

Complex multiplication is not an arbitrary two-channel real linear map. Multiplying $(x + iy)$ by $(a + ib)$ gives

$$(a + ib)(x + iy) = (ax - by) + i(ay + bx),$$

which is the real matrix

$$\begin{bmatrix} a & -b \\ b & a \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}.$$

This matrix is a rotation-and-scale coupling between the two real channels. A generic real network can learn any 2×2 channel mix, but the complex layer starts with this structured coupling built in. The coupling is not arbitrary: as Proposition 3 makes precise, complex layers are exactly the $U(1)$ -equivariant subspace of stacked-real 2-channel layers, parameterized by 2 real numbers per tap instead of 4. That can help when the task respects phase geometry, and it can hurt or be irrelevant when the task is amplitude-only or when a hand-chosen real coordinate system already exposes the right variables.

The same distinction matters for invariance. If every input is rotated by a common phase, $z \mapsto e^{i\varphi} z$, a bias-free complex linear or convolutional layer rotates its output by the same phase. This is equivariance, not magic invariance: biases, nonlinearities, readout choices, and training data determine whether the final model is actually robust to an unseen phase convention. Our fixed-rotation stress tests are designed to catch this failure mode.

1.3 Differentiation in the complex plane: Wirtinger calculus and an activation trilemma

A common worry when first building a CVNN is that backpropagation seems to require holomorphic activations, while Liouville’s theorem forbids any bounded entire nonlinearity. Both halves of that worry are imprecise. The textbook treatment in Hirose [11] and the more recent deep-CVNN survey in Trabelsi et al. [23] resolve the issue operationally by adopting split or magnitude-gated activations; we restate the resolution here in the form most useful for the rest of the paper. Backpropagation requires neither holomorphy nor even complex-differentiability; what it requires is the standard Wirtinger (CR-) calculus, in which every $\mathbb{R}$ -differentiable map $\mathbb{C} \rightarrow \mathbb{C}$ has a well-defined gradient. The genuine constraint Liouville imposes is then sharper than “no good activations exist”: it is a three-way trade-off between boundedness, holomorphy, and nonconstancy, with phase equivariance as a fourth desideratum that further constrains the design space. This subsection makes both points precise, since they govern the rest of the paper.

Wirtinger derivatives. For $f : \mathbb{C} \rightarrow \mathbb{C}$ written in real coordinates as $f(z) = u(x, y) + iv(x, y)$ with $z = x + iy$, define

$$\partial_z f = \frac{1}{2}(\partial_x f - i \partial_y f), \quad \partial_{\bar{z}} f = \frac{1}{2}(\partial_x f + i \partial_y f).$$

Both derivatives exist whenever f is real-differentiable as a map $\mathbb{R}^2 \rightarrow \mathbb{R}^2$. The Cauchy–Riemann equations are exactly $\partial_{\bar{z}} f \equiv 0$, so f is holomorphic on an open set iff $\partial_{\bar{z}} f$ vanishes there. Holomorphy is a *constraint* on f , not a prerequisite for differentiating it.

Backpropagation does not require holomorphy. Let $L : \mathbb{C}^n \rightarrow \mathbb{R}$ be a real-valued loss and let $w \in \mathbb{C}$ be a parameter. Treating L as a function of the real and imaginary parts of w , the steepest-descent direction in the underlying $\mathbb{R}^2$ corresponds to the conjugate Wirtinger gradient

$$\nabla_w L := 2 \overline{\partial_w L} = 2 \partial_{\bar{w}} L,$$

and the gradient update is $w \leftarrow w - \eta \partial_{\bar{w}} L$ (see [4, 12] for derivations). PyTorch’s complex auto-grad stores $\partial_{\bar{w}} L$ in the `.grad` field of complex parameters; training proceeds with no holomorphy assumption. Activations such as CReLU or ZReLU, which are nowhere holomorphic, are nevertheless mathematically respectable inside this calculus: they are real-differentiable almost everywhere, and their Wirtinger derivatives exist in the same a.e. sense in which ReLU' exists on $\mathbb{R}$.

The Liouville activation trilemma. The actual restriction Liouville imposes is the following.

Proposition 1 (Activation trilemma). *No function $\sigma : \mathbb{C} \rightarrow \mathbb{C}$ can simultaneously be (i) bounded on all of $\mathbb{C}$, (ii) entire (holomorphic on all of $\mathbb{C}$), and (iii) nonconstant.*

Proof. Liouville’s theorem: a bounded entire function on $\mathbb{C}$ is constant. □ □

A useful complex activation must therefore drop at least one of (i)–(iii). Each existing activation makes a different choice, and the choice is the design variable rather than an inconvenience to be hidden:

- **ComplexTanh** ($\sigma(z) = \tanh z$) keeps holomorphy and nonconstancy but drops boundedness: it is meromorphic on $\mathbb{C}$ with simple poles at $z = i(\pi/2 + k\pi)$, $k \in \mathbb{Z}$, and blows up arbitrarily close to those poles.
- **Siglog** ($\sigma(z) = z/(1 + |z|)$) keeps boundedness and nonconstancy but drops holomorphy globally: it depends on $|z|$, so $\partial_{\bar{z}} \sigma \neq 0$ on any open set.
- **CReLU**, **ZReLU**, **ModReLU**, and **ComplexCardioid** all drop holomorphy outright. They are $\mathbb{R}$ -differentiable a.e., bounded only on bounded subsets of $\mathbb{C}$, and chosen for their interaction with magnitude and phase rather than for any analytic property.

A fourth desideratum: phase equivariance. For phase-bearing data the natural symmetry is the $U(1)$ action $z \mapsto e^{i\varphi} z$. An activation respects this symmetry iff

$$\sigma(e^{i\varphi} z) = e^{i\varphi} \sigma(z) \quad \forall \varphi \in \mathbb{R}, z \in \mathbb{C}. \quad (1)$$

Combining (1) with holomorphy is more restrictive than Liouville alone:

Proposition 2 (Phase-equivariant entire functions are linear). *If $\sigma : \mathbb{C} \rightarrow \mathbb{C}$ is entire and satisfies (1), then $\sigma(z) = cz$ for some $c \in \mathbb{C}$.*

Proof. Expand $\sigma(z) = \sum_{n \geq 0} a_n z^n$ on $\mathbb{C}$. Equation (1) gives $\sum_n a_n e^{in\varphi} z^n = \sum_n a_n e^{i\varphi} z^n$ for all φ , hence $a_n(e^{in\varphi} - e^{i\varphi}) = 0$ for all $\varphi \in \mathbb{R}$, which forces $a_n = 0$ for $n \neq 1$. □ □

So the four desiderata — bounded, holomorphic, nonconstant, phase-equivariant — are jointly satisfiable only by the trivial constant or linear maps. The six activations characterized in Section 3.1 can be read as a natural experiment over this $2 \times 2 \times 2 \times 2$ taxonomy with the useless cells removed: **ModReLU** and **Siglog** keep phase equivariance at the cost of holomorphy; **ComplexTanh** keeps holomorphy at the cost of boundedness; **CReLU**, **ZReLU**, and **ComplexCardioid** drop both holomorphy and equivariance to buy other useful properties (split-channel sparsity, quadrant selection, magnitude gating).

The CR-residual is the trilemma made measurable. The Cauchy–Riemann residual reported in Figure 1 is, up to a factor of two, exactly $|\partial_{\bar{z}} \sigma|$ evaluated on a grid in $\mathbb{C}$. It is zero on open sets where σ is holomorphic and strictly positive elsewhere, so it quantifies the deviation from holomorphy that Proposition 1 forces every nonconstant bounded activation to incur. Pairing CR-residual against gradient norm at initialization turns the abstract trilemma into a per-activation diagnostic: an activation either pays a holomorphy cost (positive CR residual) or a boundedness cost (large intrinsic Jacobian), and the figure shows each candidate’s chosen trade.

With this calculus in place, the rest of the paper treats activation choice as a structured design decision rather than a workaround for a theorem that does not actually obstruct backpropagation.

1.4 What this paper contributes

The paper makes three contributions at a scale where the mechanisms can be inspected:

- **A representation-first evaluation protocol.** We compare native complex models against real Cartesian, parameter-matched real, FLOP-matched real, polar, phase-only, and magnitude-only baselines. This separates “complex arithmetic helped” from “the model was simply given the right coordinates.”
- **Stress tests across domains.** Synthetic RF, quantum wavefunctions, and EEG analytic signals show the same pattern: phase-bearing tasks reward phase-aware views; amplitude-heavy tasks reward magnitude; coupled phase/amplitude tasks reward joint representations; and unseen phase conventions require explicit testing or augmentation.
- **An optimization caution for RadioML.** On a 3-class RadioML subset, a large matched-shared-trial gap is mostly a high-learning-rate robustness artifact. Independent per-family tuning gives a much smaller gap, and gradient telemetry explains why.

The paper does *not* claim:

- That complex arithmetic is universally better than real-valued learning. The RF stress tests deliberately include cases where real magnitude or matched real baselines win. The contribution is to map the boundary of the advantage, not to erase the boundary.
- That CVNNs beat real-valued networks at full RadioML 2018.01A scale (24 modulations $\times$ 26 SNR levels $\times$ 1024-sample length). The infrastructure for that experiment is in place (`experiments/rf/sweep_radioml.py -preset full`), but the run is deferred. The mechanism analysis is the contribution; scale-up is signposted future work.
- That the inductive bias survives in 2-D regimes (audio STFT, MRI). `ComplexConv2d` is not implemented; the project is 1-D only.
- That the quantum pilot establishes broad physics superiority. The quantum and EEG pilots are deliberately small application probes, included to test whether the representation thesis transfers outside RF.
- That CReLU activations generalize across signal-processing settings. The activation ablation (Section 3.8.1) is part of the contribution precisely because activation choice changes both the apparent size and, in one row, the direction of CVNN-vs-real gaps.

2 Experimental design

Section 1 turns the motivation into three empirical questions: what information is present, what algebraic constraint the model imposes, and how optimization changes under that constraint. The experimental design keeps these questions separate.

Coordinate views. For any complex sequence $z_t = x_t + iy_t$, the same sample can be presented as Cartesian channels (x_t, y_t) , polar channels $(|z_t|, \cos \theta_t, \sin \theta_t)$, phase-only channels $(\cos \theta_t, \sin \theta_t)$, or magnitude-only $|z_t|$. These views are not weaker or stronger in a single ordering. They encode different hypotheses: phase-only should win when labels depend on angle or phase lag; magnitude-only should win when labels depend on energy; polar should win when amplitude and phase interact; native complex layers should help when the rotation-and-scale coupling of complex multiplication matches the task geometry.

Model families. Each benchmark compares the native complex network with real-valued baselines that control for different confounds: a Cartesian real network on stacked real/imaginary channels, a parameter-matched real network, a FLOP-matched real network, and a rotation-equivariant real reparameterization that constrains each 2-channel kernel tap to the $aI + bJ$ subspace of $M_2(\mathbb{R})$ identified by Proposition 3. The rotation-equivariant baseline is provably equivalent to the complex layer (forward and gradient, verified in `tests/test_rotation_equivariant.py`); we use it as a correctness witness rather than a separate experimental row, and discuss why the substantive comparison is therefore complex versus the unconstrained Cartesian and parameter-matched baselines in Section 3.4. Architecture details and parameter formulas are deferred to Appendix C; the main

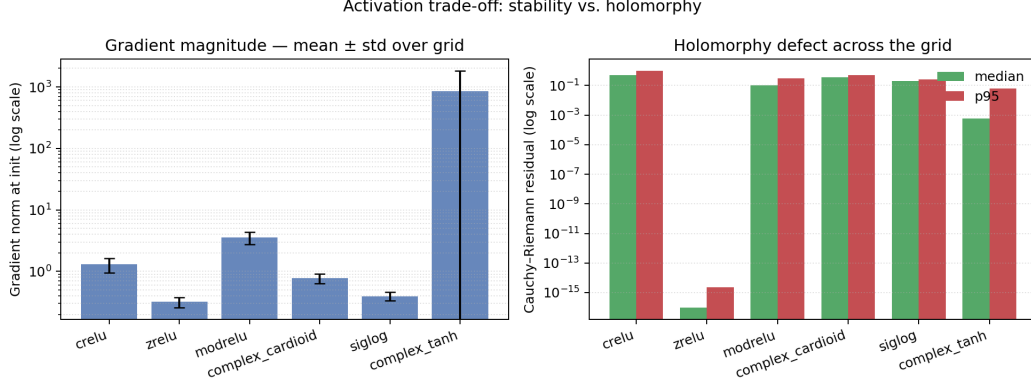


Figure 1: Liouville trade-off, made empirical. *Left*: gradient norm at init for each candidate activation, log scale. *Right*: median and p95 Cauchy–Riemann residual on a $[-3, 3]^2$ grid in $\mathbb{C}$, log scale. `complex_tanh` is locally holomorphic but unbounded (poles at $\pm i\pi/2$ inside the grid); `siglog` is bounded but never holomorphic; `crelu`, `zrelu`, `modrelu`, and `complex_cardioid` sit between, each making a different compromise.

point is that every reported win must survive at least one plausible real-valued way of seeing the same information.

Nonlinearity in the complex plane. Activation choice is a first-principles issue rather than a software detail. Section 1.3 sets up the Wirtinger calculus that makes complex backpropagation well-defined and states the trilemma (Proposition 1) any nonconstant complex activation must navigate; in this section we use that taxonomy to fix the candidate set. We characterize six activations before using them downstream: CReLU, ZReLU, ModReLU [2], ComplexCardioid, Siglog, and ComplexTanh. Together they cover the non-degenerate cells of the trilemma — bounded but non-holomorphic, holomorphic but unbounded, and split-channel constructions that drop both holomorphy and phase equivariance to gain other properties.

Stress-test factors. The RF synthetic suite varies the data-generating mechanism rather than only the model: PSK-only, QAM-only, mixed PSK+QAM, low and high SNR, unit-magnitude normalization, unit-power normalization, fixed carrier-phase rotation, random-rotation augmentation, and activation choice. The quantum and EEG pilots reuse the same logic in other domains by constructing tasks where the label is known to live in phase, magnitude, or a phase-amplitude relation.

Selection rules. Every model family draws from the same search space and consumes the same trial-by-seed budget. The matched-shared-trial rule chooses the complex reference trial by validation accuracy, then reports every baseline at that same trial index; this is a stress test for robustness under a shared hyperparameter allocation. Independent per-family selection is also reported where it changes the interpretation, because it estimates best-achievable performance when each family is tuned separately.

3 Experiments and results

3.1 Activation characterization

The activation question is where complex-valued neural networks first depart from the intuition of ordinary real-valued networks. A useful activation must be nonlinear; a globally bounded holomorphic function cannot be nonlinear. Practical CVNN activations therefore choose a compromise. Some are locally analytic but can blow up; some are bounded but explicitly non-holomorphic; some preserve global phase rotations while others deliberately break them.

Figure 1 makes the Liouville bind empirically visible. `complex_tanh` has near-zero CR residual at the median — it really is locally holomorphic — but the same metric’s max is 905, because the grid

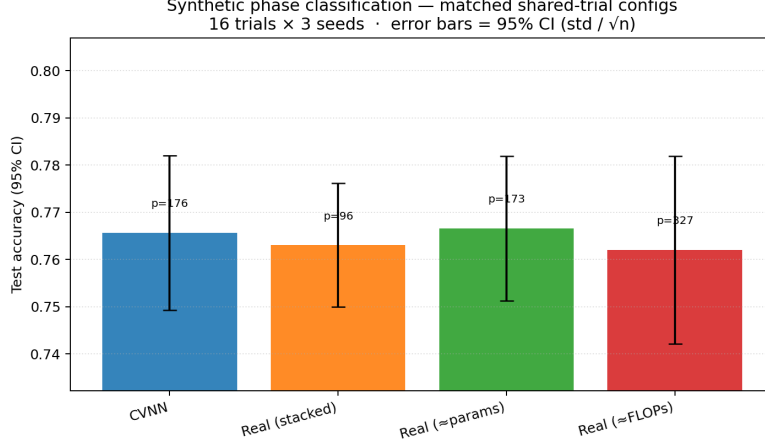


Figure 2: 1-D phase classification: matched shared-trial test accuracy after a 16×3 sweep. All four families converge to $\sim 76\%$ with overlapping CIs — the complex inductive bias has nothing to bite on at this scope.

(extent = 3) brackets the first pair of poles at $z = \pm i\pi/2 \approx \pm 1.5708i$. We emphasize that this large residual is not a Cauchy–Riemann violation in the analytic sense: $\tanh$ is holomorphic at every non-pole point, with $\partial_{\bar{z}} \tanh \equiv 0$ exactly there. The plotted residual is a finite-difference estimate of $|\partial_{\bar{z}} \tanh|$, and finite differences inflate near singularities. The plot is honest as an *operational* diagnostic of activation behaviour on a fixed grid (which is what training sees), but the trilemma cell `complex_tanh` occupies is “meromorphic, unbounded” rather than “never holomorphic.” The intrinsic Jacobian mean is ~ 108 . By contrast, `siglog` is bounded ($|f| < 1$ everywhere) and never blows up, but its CR residual is everywhere positive ($p95 = 0.25$) — it is not holomorphic anywhere, by construction. The remaining four activations sit between, each making a different compromise.

This is not a finding about which activation is best in training — it is a visible measurement of the formal trade-off the project is built around. The choice of activation in the downstream benchmarks (default `crelu`, following the pragmatic choice in Trabelsi et al. [23]) is consistent across runs and isolated from the architectural variables. Our results in Section 3.8.4 and Section 3.8.6 qualify the `crelu` default in a way Trabelsi et al. did not have to contend with: `crelu` interacts asymmetrically with matched-shared-trial hyperparameter selection through its split-ReLU readout sensitivity, while `zrelu` and `modrelu` (the latter from Arjovsky et al. [2] and extended by Wisdom et al. [27] as the unitary-RNN nonlinearity) do not. The activation choice and the selection rule cannot be discussed independently.

3.2 Synthetic phase classification — null result

After a 16×3 budgeted sweep, the matched shared-trial comparison puts all four families at ~ 76.2 – 76.7% test accuracy with overlapping CIs (Figure 2). On 1-D phase classification, **the complex inductive bias does not help**. A 2-D real network on (Re, Im) already has access to all the information the task contains; there is no temporal or geometric structure for $\mathbb{C}$ -multiplication to exploit. This is exactly what the phase-equivariance analysis in Section 4.1 predicts: a single-sample phase task does not invoke any sequence-level equivariance.

3.3 Synthetic RF modulation — complex wins under matched budget

Table 1 reports the matched shared-trial comparison after a $16\text{-trial} \times 6\text{-seed}$ sweep with the conv architecture (`ComplexConv1d` $\rightarrow$ activation $\rightarrow$ `ComplexConv1d` $\rightarrow$ activation $\rightarrow$ global mean pool $\rightarrow$ `ComplexLinear` $\rightarrow |\cdot|$). Complex beats every real baseline by ≥ 3.8 percentage points. Because the matched-shared-trial rule evaluates all four families on the same six seeds, we use a *paired t*-test on per-seed differences ($df = 5$) rather than the unpaired Welch approximation: pairing removes between-seed variance and is the correct test for this design. The paired statistics are $t = 4.44$ vs. stacked, $t = 6.33$ vs. matched-parameter ($p \approx 0.0014$), and $t = 8.89$ vs. matched-FLOP

Table 1: Synthetic RF modulation classification, matched shared-trial selection after a 16×6 budgeted sweep with the conv architecture ($n = 6$ seeds). Per-family 95% CIs use the t -distribution ($t_{0.975,5} = 2.571$); the rightmost column reports the paired difference complex – baseline with its 95% paired- t CI on per-seed differences. All three paired differences exclude zero with margin.

family	test acc	std	params	95% CI ($t_{0.975,5}$)	paired Δ vs. complex
complex	0.8191	0.0163	3,974	[0.8020, 0.8362]	—
real (stacked)	0.7740	0.0226	2,099	[0.7503, 0.7977]	+0.0452 [+0.019, +0.071]
real ($\approx$ params)	0.7769	0.0187	3,809	[0.7573, 0.7965]	+0.0423 [+0.025, +0.059]
real ($\approx$ FLOPs)	0.7810	0.0195	7,779	[0.7606, 0.8015]	+0.0382 [+0.027, +0.049]

($p \approx 0.0003$); all three paired-difference 95% CIs exclude zero. The per-family t -CIs in the table also exclude overlap with complex for the stacked and matched-parameter baselines and touch the complex CI at the boundary for matched-FLOP — the paired test, not the per-family CIs, is the load-bearing inference under matched seeds.

3.4 Why no rotation-equivariant baseline row

Proposition 3 characterizes complex layers as exactly the $U(1)$ -equivariant subspace of stacked-real 2-channel layers. Operationally this means the natural “complex inductive bias, implemented in real coordinates” baseline — a real Conv1d whose kernel taps are constrained to $aI + bJ$, with channel-wise ReLU and a magnitude readout — is not a separate model; it is the same model as ComplexConv1d with CReLU and $|\cdot|$ readout, modulo the channel layout convention.

We make this concrete in `cvnn.baselines.rotation_equivariant` and the test suite `tests/test_rotation_equivariant.py`, which asserts numerical equivalence of the forward pass and per-parameter gradients under matched initialization ($\text{atol} \leq 10^{-6}$) and verifies that three AdamW optimization steps from a shared self-supervised loss leave (a, b) within 1.5×10^{-8} of $(\text{Re}(W), \text{Im}(W))$ — the difference between float32 representations of the same trajectory. The empirical “rotation-equivariant baseline ablation” would therefore report identical numbers to the complex row, by construction. We elect to report the equivalence as a tested theorem rather than a redundant table row.

The substantive comparison in Table 1 is therefore complex (equivalently, rotation-equivariant 2-parameter-per-tap real) *versus the unconstrained 4-parameter-per-tap stacked, matched, and FLOP-matched real baselines*. The 3.8–4.5 pp matched-shared-trial gap is the cost the unconstrained baselines pay for spending the extra two parameters per tap on the asymmetric complement of $\iota(\mathbb{C})$ — parameters that, if the task respects $U(1)$, are at best wasted and at worst overfit the asymmetry. Read this way, the win is not “complex arithmetic helped”; it is “the $U(1)$ -equivariant 2-parameter subspace generalizes better than its 4-parameter superset on PSK-like data,” which is exactly what Proposition 3 predicts.

3.5 RF representation stress tests — where the claim breaks

The matched-budget result in Section 3.3 answers one question: under a conventional Cartesian real baseline, does the complex conv stack help? It does. But it does not answer the more important mechanism question: is the advantage due to native complex arithmetic, phase information, amplitude information, or a coordinate choice that a real network could also receive? The representation stress tests in Figure 3 make those alternatives explicit.

Three conclusions follow. First, PSK-only modulation is the cleanest case for the complex prior: magnitude-only is exactly chance while phase-aware real views and complex models learn. Second, the same statement is false for QAM-only data: amplitude carries class structure, and magnitude-only is the best single view in the committed stress test. Third, native complex layers are not automatically invariant to a carrier-phase offset: fixed rotation at test time drops all coordinate-dependent PSK models to chance-level behavior, and random rotation augmentation only partially restores performance. This is the key scope refinement. The paper should not claim “complex beats real” as a universal result; it should claim that complex-valued models are a useful inductive bias for RF signal learning when the task’s phase/amplitude geometry and invariance assumptions match the model.

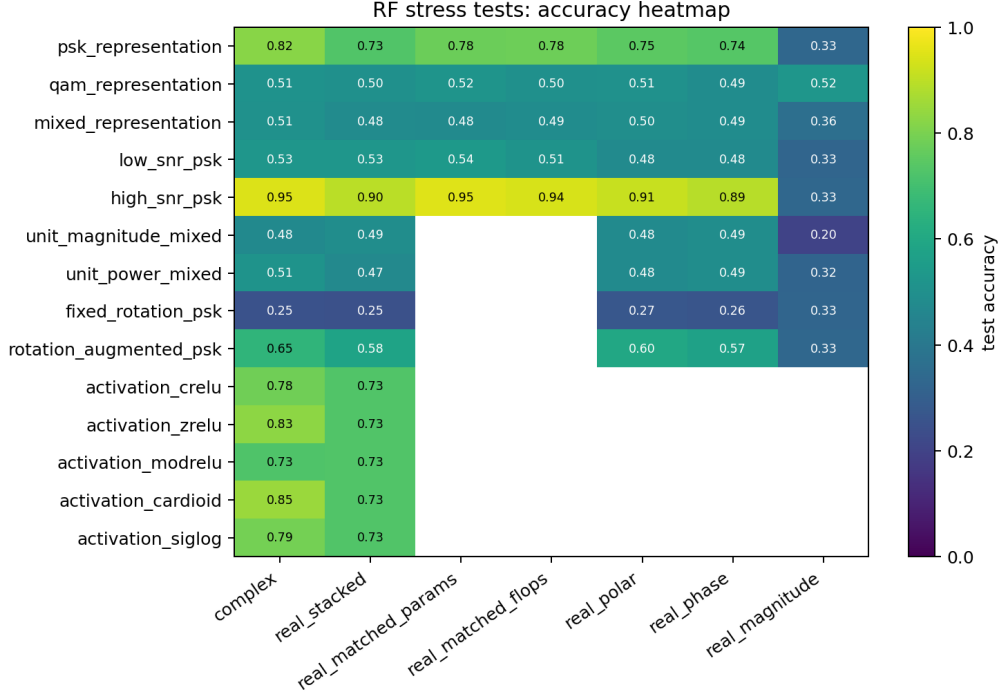


Figure 3: RF synthetic representation stress tests. Rows vary the modulation family, SNR regime, normalization, rotation shift, and complex activation; columns vary the model or real coordinate representation. The heatmap is useful precisely because it shows both wins and failures: complex wins PSK and mixed regimes, but magnitude wins QAM-only; fixed rotation collapses every coordinate-dependent model.

Table 2: Selected RF synthetic stress-test outcomes. Accuracy is mean test accuracy across the committed stress-test seeds. The central lesson is conditionality: PSK rewards phase-aware geometry, QAM rewards magnitude, low SNR weakens phase-only views, and global carrier-phase shifts require augmentation or an explicit invariance mechanism.

condition	best	complex	real stack	phase	magnitude
PSK-only	complex 0.821	0.821	0.728	0.735	0.333
QAM-only	magnitude 0.524	0.509	0.502	0.487	0.524
mixed PSK+QAM	complex 0.507	0.507	0.481	0.487	0.365
low-SNR PSK	matched real 0.537	0.526	0.526	0.479	0.326
high-SNR PSK	matched real 0.952	0.949	0.897	0.892	0.333
unit-magnitude mixed	phase 0.491	0.476	0.490	0.491	0.200
fixed-rotation PSK	magnitude 0.328	0.252	0.246	0.262	0.328
rotation-augmented PSK	complex 0.654	0.654	0.580	0.574	0.333

3.6 Quantum wavefunction pilot — physics application

To test whether the same representation story transfers outside RF, we added a small physics pilot over 1-D complex wavefunctions. The data are synthetic but physically grounded. In the first task, examples are Gaussian wavepackets $\psi(x) = A(x) \exp(ikx + i\phi_0)$ and the model predicts one of four momentum classes. The amplitude envelope is randomized independently of class, so the class information is in the phase gradient rather than in $|\psi|$. In the second task, a right-moving wavepacket is evolved under the time-dependent Schrödinger equation with a split-step Fourier integrator, and the model predicts one of five potential families: free, barrier, well, double barrier, or harmonic. We

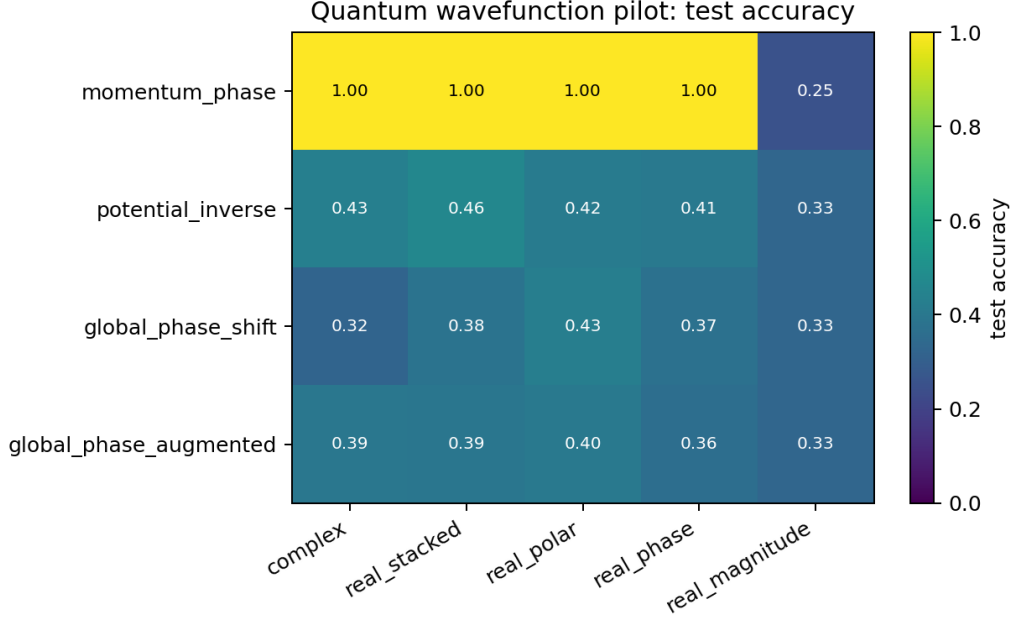


Figure 4: 1-D quantum wavefunction pilot. The momentum task stores the label in the phase gradient of $\psi(x)$, so magnitude-only stays at chance while phase-aware views solve it. The potential-inverse tasks are above chance but do not give a universal complex win: explicit real coordinate systems match or beat the complex model, and global-phase robustness must be tested directly.

Table 3: Quantum wavefunction pilot, standard preset (3 seeds, 128 examples/class, 96 grid points, 140 training steps). Chance is 0.25 for momentum and 0.20 for potential classification. The result supports a physics application of the representation thesis, not a broad complex-dominance claim.

condition	best	complex	real stack	phase	polar	magnitude
momentum phase	complex 1.000	1.000	1.000	1.000	1.000	0.250
potential inverse	real stack 0.462	0.431	0.462	0.405	0.418	0.331
global phase shift	polar 0.428	0.323	0.382	0.374	0.428	0.331
global phase augmented	polar 0.403	0.395	0.392	0.362	0.403	0.331

also evaluate the potential task under an unseen global phase rotation $\psi \mapsto i\psi$ and with random global-phase augmentation at train time.

Table 3 gives the useful scope boundary. First, phase can be indispensable in a physics state: all phase-aware or Cartesian views reach 1.000 on momentum classification, while magnitude-only is exactly chance. This is the quantum analogue of the PSK result. Second, the potential-inverse task is learnable above chance, but the best model is the explicit real Cartesian baseline (0.462) rather than the complex model (0.431). Third, global phase is a physical nuisance symmetry, but the current complex conv stack does not automatically solve it: under an unseen $\pi/2$ phase shift, complex drops to 0.323 while magnitude remains unchanged at 0.331. Random global-phase augmentation partially recovers the complex model (0.395), but does not make this a solved invariance problem. The physics pilot therefore strengthens the paper’s positioning: complex models are plausible for domains with native complex fields, but the right claim is still representation- and symmetry-conditional.

3.7 EEG analytic-signal pilot — unexpected biomedical application

As a deliberately unexpected application, we added a synthetic EEG/MEG-style analytic-signal pilot. In many biomedical signal pipelines, a real sensor trace is mapped through a Hilbert or wavelet

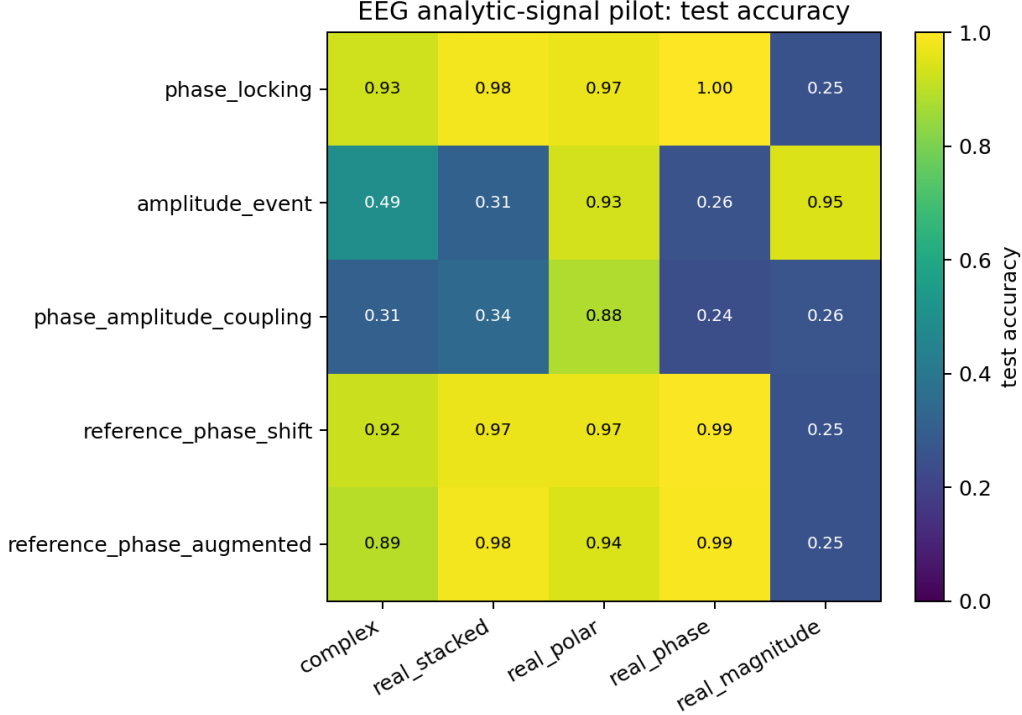


Figure 5: Synthetic EEG analytic-signal pilot. Rows separate three common oscillatory mechanisms: inter-channel phase locking, amplitude bursts, and phase-amplitude coupling. The winning representation changes with the mechanism: phase-only solves phase locking, magnitude-only solves amplitude events, and polar coordinates solve phase-amplitude coupling.

Table 4: EEG analytic-signal pilot, standard preset (3 seeds, 128 examples/class, 4 channels, 64 time steps, 120 training steps, `crelu`). Chance is 0.25. The result is a representation map, not a claim about clinical EEG performance.

condition	best	complex	real stack	phase	polar	magnitude
phase locking	phase 1.000	0.926	0.981	1.000	0.971	0.250
amplitude event	magnitude 0.946	0.494	0.311	0.256	0.933	0.946
phase-amplitude coupling	polar 0.881	0.308	0.343	0.240	0.881	0.263
reference phase shift	phase 0.994	0.920	0.971	0.994	0.971	0.250
reference phase augmented	phase 0.990	0.891	0.984	0.990	0.942	0.250

transform into a complex analytic signal whose magnitude is an amplitude envelope and whose angle is instantaneous phase. This makes neuroscience a natural but non-obvious test of the paper’s claim: does the complex dtype help, or does the relevant coordinate view matter more?

Table 4 makes the application boundary concrete. For phase locking, the class is the phase lag between two channels; phase features solve it and magnitude-only is exactly chance. For amplitude events, the class is the channel containing an amplitude burst; magnitude and polar views solve it while phase-only is chance. For phase-amplitude coupling, the class is a relationship between a low-frequency phase and a high-frequency amplitude envelope; neither phase-only nor magnitude-only is enough, while polar coordinates reach 0.881. Native complex models are competitive on phase-locking but do not dominate the pilot. This is still useful for the paper: it shows that the same stress-test protocol can explain an unexpected biomedical domain without relying on a blanket “complex is better” story.

Table 5: RadioML 2018.01A subset, CReLU run, matched shared-trial selection after 16×6 sweep. Read in isolation, a dramatic CVNN-vs-real win at matched parameters — but see Section 3.8.1 and Section 3.8.4.

family	test acc	std	params
complex (CReLU)	0.7293	0.0085	58,886
real (stacked)	0.4583	0.1394	29,891
real ($\approx$ params)	0.4999	0.1327	58,413
real ($\approx$ FLOPs)	0.4245	0.1415	117,123

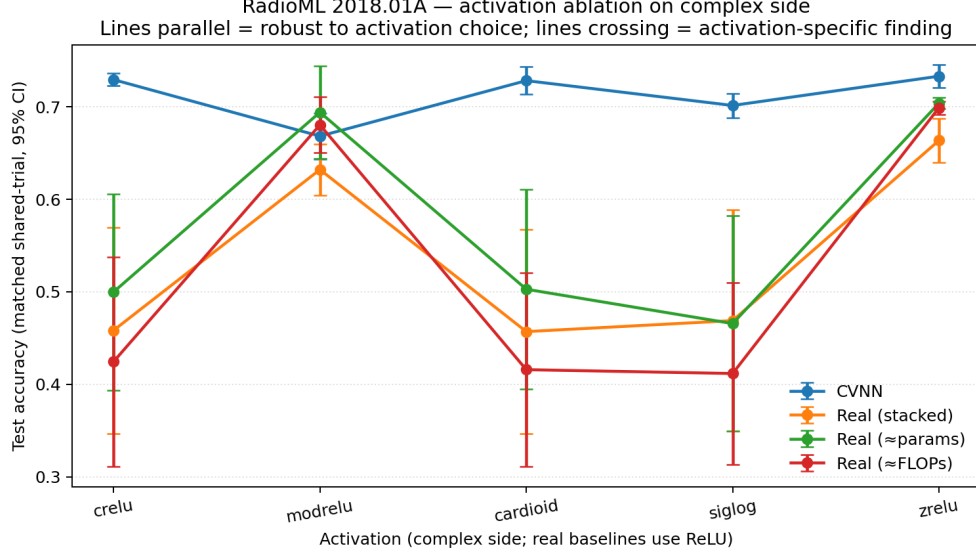


Figure 6: Five-activation ablation on the same RadioML scaffold. The CVNN line is nearly flat across all five activations (test accuracy 0.668–0.733); real-baseline lines swing wildly between ~ 0.45 and ~ 0.70 depending on which activation the complex side uses. The 23 pp gap from Table 5 is mostly real baselines collapsing under the matched-shared-trial selected hyperparameters, not complex pulling ahead.

3.8 RadioML 2018.01A — a positive result with a subtler story

The real-data RF study uses a controlled 3-class RadioML 2018.01A subset: BPSK / QPSK / 8PSK at $\{-10, -6, -2, 2, 6, 10, 14, 18\}$ dB, with `max_per_class_per_snr=256` and `sample_length=128`. This subset is intentionally narrower than the full archive. It keeps the task phase-dominant enough to compare against the synthetic PSK stress tests, while remaining large enough to expose real training dynamics. A strict loader validates the even-SNR convention used by RadioML 2018.01A; details are in Appendix G. The CReLU run on an NVIDIA A100 produced a striking 23 pp gap:

Read in isolation, Table 5 looks like a dramatic complex-vs-real win at matched parameters — and an earlier draft claimed exactly that. The activation ablation in Section 3.8.1 forces a different reading.

3.8.1 Activation ablation reveals a methodology-driven artifact

We re-ran the same sweep four more times, varying only the complex side’s activation; the real baselines always use ReLU. The matched-shared-trial selection per activation (Table 6) shows complex’s accuracy nearly flat (range 0.065) while real baselines swing by 0.27. Figure 6 makes the asymmetry visible. The 22–23 pp gap on `crelu` / `cardioid` / `siglog` is *mostly the real baselines failing under matched-shared-trial selection*, not the complex network pulling ahead. The robustness asymmetry

Table 6: Five-activation ablation, matched shared-trial test accuracy. Under stable activations (modrelu, zrelu) real baselines are also stable and the gap drops to ± 3 pp; under unstable activations the gap inflates. See Section 3.8.4 for the mechanism.

activation	complex	best real	gap (pp)	$\sigma(\text{complex})$	mean $\sigma(\text{real})$
crelu	0.7293	0.4999	+22.94	0.009	0.138
cardioid	0.7282	0.5028	+22.54	0.019	0.134
siglog	0.7014	0.4689	+23.25	0.016	0.139
modrelu	0.6683	0.6940	-2.58	0.031	0.045
zrelu	0.7330	0.7039	+2.91	0.015	0.015

is the durable, defensible finding; the apparently large per-activation gap is a consequence of that asymmetry, not a separate phenomenon.

3.8.2 Per-SNR breakdown

At -10 dB everyone is near chance ($\sim 33\%$ for 3 classes); from $+2$ dB onward, the CReLU complex network reaches 92.1% at $+10$ dB and 91.6% at $+18$ dB while the real baselines plateau at 47–61%. This is the same pattern as Table 5: the gap reflects real baselines failing to train at the selected hyperparameters under CReLU, not complex pulling cleanly away under every stable activation regime.

3.8.3 Comparison to published RadioML numbers

The RadioML 2018.01A dataset paper [18] reports a real-valued ResNet-style classifier reaching roughly 90% top-1 at high SNR on the full 24-class task with the full sample length 1024. Subsequent CVNN-on-RadioML work [13, 24] reports a typical complex-vs-real advantage of a few percentage points on the same 24-class setup with comparable architectures. Our +3-and-flat ablation result on modrelu / zrelu (the activations under which real baselines are also stable) is consistent with that small reported gap. The 22–23 pp gap on crelu / cardioid / siglog is *not* a contradiction of the literature — it is the matched-shared-trial selection rule magnifying real-baseline instability that the literature’s separate-tuning protocols would not expose.

3.8.4 Mechanism — explosion-into-dead-region at step 1

Section 3.8.1 reports an asymmetry without explaining it. We re-ran the matched-shared-trial selected configuration with per-step instrumentation (train loss, total parameter gradient norm, per-parameter gradient norm, max parameter magnitude) for each of $5 \times 4 \times 3 = 60$ runs, capped at 200 steps (the divergence pattern appears within the first 5).

The mechanism is unambiguous (Figure 8). On crelu / cardioid / siglog, matched-shared-trial picks a high-learning-rate / wide-hidden config that complex selects because it tolerates it (learning rate $\in \{0.024, 0.024, 0.040\}$, hidden = 64). At step 1, every family sees a large AdamW step away from random init. For real baselines this produces a step-1 train loss in $[3, 36]$, a total gradient norm in $[6, 42]$, and a head.weight gradient in $[6, 41]$ — i.e. nearly all of the explosion sits on the final classifier’s weights. For the complex network the same step produces a step-1 loss in $[1.07, 1.28]$, total gradient norm $[0.3, 1.0]$, head.weight gradient $[0.1, 0.7]$ — about 50–80 $\times$ smaller than its real counterparts on identical data and learning rate.

The complex network’s (Re, Im) parameter coupling distributes the loss signal across more parameter slots, so the same $\text{lr} \times \text{CE-gradient}$ product produces a smaller effective step on head.weight. The real network does not have that smoothing; AdamW’s first step pushes some seeds into a region where the second Conv1d’s ReLU activations are all-zero (or all-active and saturated), gradients vanish, and training “stabilizes” at uniform-prediction chance level. We confirmed this: *the dead seeds end at exactly* loss = $\ln 3 \approx 1.099$ (3-class CE under uniform predictions), with total_grad_norm ≤ 0.05 , and zero test-accuracy improvement from random init.

Table 7 summarizes the dead-seed counts. **The dead-seed rate jumps from 0% to 33% as soon as the matched-shared-trial selected learning rate crosses ~ 0.02 .** Two orders of magnitude separation in the step-1 head gradient between the unstable and stable activation regimes.

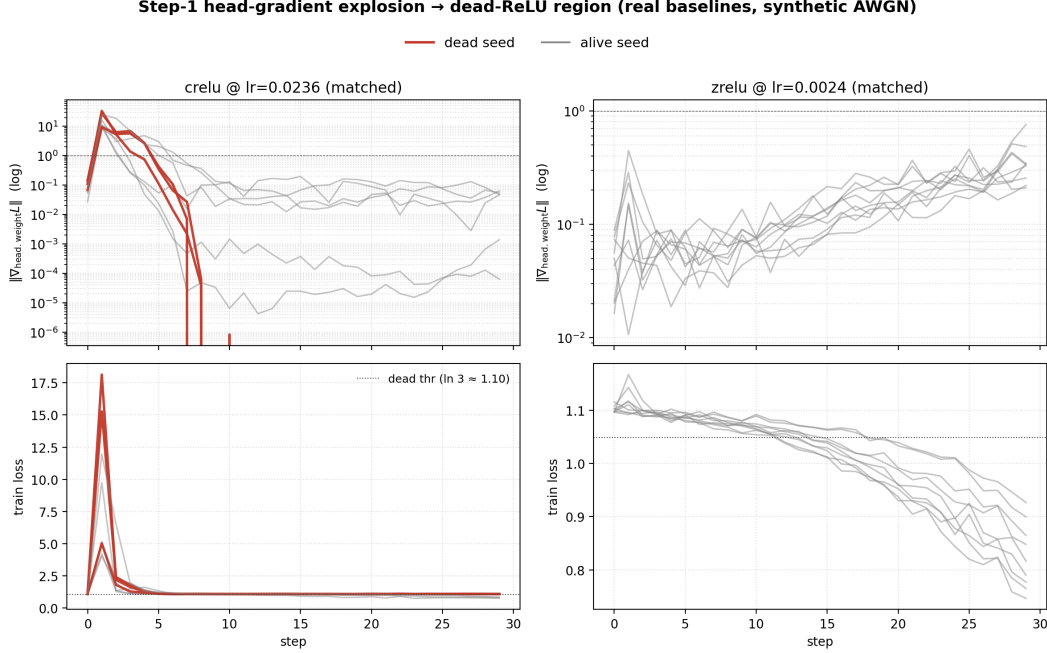


Figure 7: Per-step trajectories on synthetic AWGN. *Top row:* head.weight gradient (log scale). *Bottom row:* train loss with the 3-class chance threshold ($\ln 3$) overlaid. Under `crelu@matched` (left), three real-baseline seeds spike above $\|\nabla\| \sim 10$ at step 1, then collapse to $\sim 10^{-6}$ within five steps and never recover (loss pinned at $\ln 3$). Under `zrelu@matched` (right) the same families train smoothly. The mechanism is local to the high-learning-rate AdamW first step, not the rest of training.

Table 7: Dead-seed counts (out of 9 per activation: three families $\times$ three seeds), step-1 head.weight gradient. The threshold behavior is striking: 0% $\rightarrow$ 33% as learning rate crosses ~ 0.02 . The same pattern replicates on synthetic AWGN data (Section 3.8.5).

activation	learning rate	dead seeds	step-1 head.weight grad (real, max)
<code>crelu</code>	0.024	3/9	13.7
<code>cardioid</code>	0.024	3/9	19.5
<code>siglog</code>	0.040	3/9	40.4
<code>modrelu</code>	0.008	0/9	0.17
<code>zrelu</code>	0.0024	0/9	0.13

3.8.5 Cross-task confirmation

To check whether the explosion-into-dead-region pattern is RadioML-specific (pulse shaping, carrier offset, channel effects) or generic to the hyperparameter regime, we re-ran the same 60-run telemetry against the synthetic AWGN-only RF generator using the same per-activation hyperparameters that the RadioML matched-shared-trial selected. Table 8 shows the side-by-side dead-seed counts.

3.8.6 Learning-rate-vs-activation disambiguation

Table 7 reads as if some activations (`crelu`, `cardioid`, `siglog`) are intrinsically unstable and others (`modrelu`, `zrelu`) are intrinsically stable. A skeptical reading is that the same data is just as consistent with the dead-seed pattern being driven by the *learning rate* that matched-shared-trial selects, not the activation: the unstable rows all selected learning rates in $[0.024, 0.040]$, the stable rows in $[0.0024, 0.008]$. To disambiguate, we ran a 2×2 factorial on synthetic AWGN data: $\{\text{crelu}, \text{zrelu}\} \times \{\text{learning rate} = 0.0236, \text{learning rate} = 0.0024\}$, holding all other hyperparameters at

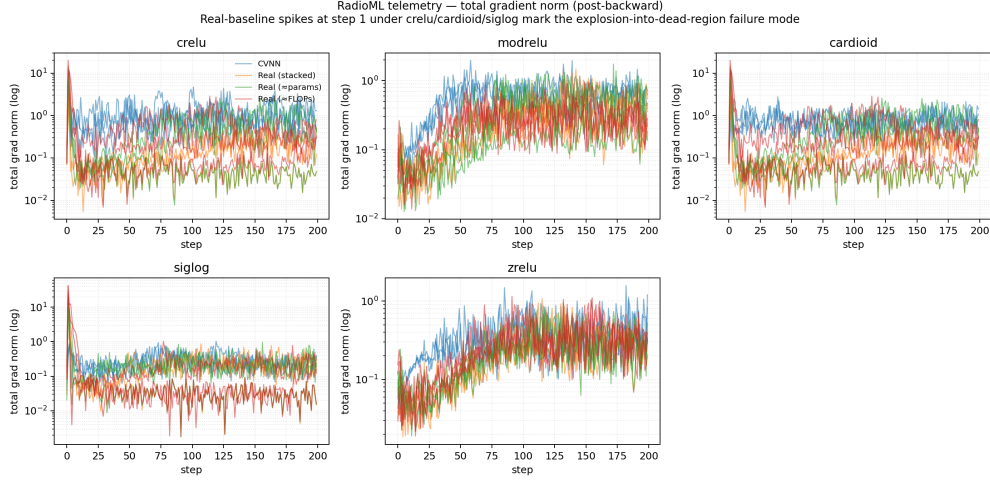


Figure 8: RadioML telemetry — total gradient norm post-backward, log y -axis, one panel per activation. Real-baseline spikes at step 1 under `crelu` / `cardioid` / `siglog` mark the explosion-into-dead-region failure mode that produces the inflated gaps in Table 6; under `modrelu` and `zrelu` no such spikes occur and the gap collapses to ± 3 pp.

Table 8: Cross-task replication: same matched-shared-trial hyperparameters applied to synthetic AWGN-only RF data. Dead-seed counts replicate almost exactly; step-1 gradients on synthetic are if anything larger than RadioML’s because AWGN-only signals carry more per-sample variance than pulse-shaped, channel-attenuated ones. *The threshold is hyperparameter-driven, not data-driven.*

activation	RadioML dead/9	synthetic dead/9	RadioML max real grad	synthetic max real grad
<code>crelu</code>	3	3	19.9	34.1
<code>cardioid</code>	3	3	19.9	34.1
<code>siglog</code>	3	2	42.1	55.7
<code>modrelu</code>	0	0	0.3	0.5
<code>zrelu</code>	0	0	0.2	0.7

each activation’s matched-shared-trial selection. Three real families $\times$ three seeds per cell, capped at 200 steps.

Table 9 reports the disambiguation. **Learning rate is the dominant axis.** Dropping `crelu` from its matched learning rate (0.0236) to `zrelu`’s learning rate (0.0024) eliminates dead seeds entirely ($4/9 \rightarrow 0/9$) and shrinks the step-1 head gradient by an order of magnitude ($31.4 \rightarrow 2.3$). Conversely, forcing `zrelu` to `crelu`’s learning rate produces a step-1 head gradient of 6.3 and the first dead seed under that activation in this study ($1/9$). Both activations cross from “stable” to “unstable” as learning rate crosses ~ 0.02 — the threshold from Table 7 is hyperparameter-driven, not activation-driven, and replicates the Section 3.8.4 mechanism with the activation factor swapped in.

Activation is a secondary modulator. At the same high learning rate, `crelu` produces a step-1 head gradient $\sim 5\times$ larger than `zrelu` (31.4 vs 6.3) and four times the dead-seed rate ($4/9$ vs $1/9$); the split-ReLU structure of `crelu` amplifies the readout’s response to a high-learning-rate first step but does not by itself trigger the failure mode at safe learning rates. The honest reading of Table 7 is therefore not “some activations are unstable and others are stable” but “the matched-shared-trial rule selected unsafe learning rates for some activations and safe ones for others, and that learning-rate difference — modulated by activation sensitivity — is what drives the dead-seed counts.”

3.9 The matched-shared-trial selection artifact

The activation ablation (Section 3.8.1), the per-step mechanism (Section 3.8.4), and the learning-rate-vs-activation factorial (Section 3.8.6) together support a methodological finding that is independent of any particular complex-vs-real comparison: *the matched-shared-trial hyperparameter*

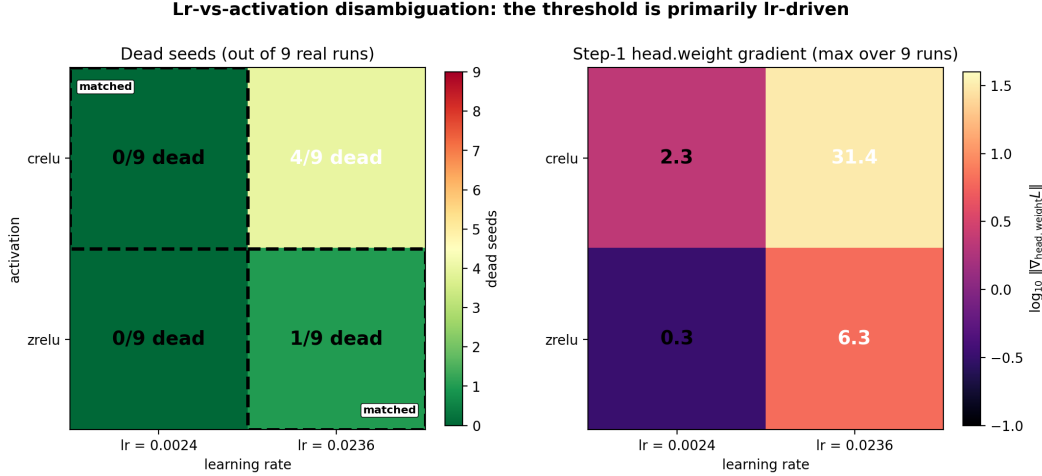


Figure 9: Learning-rate-vs-activation 2×2 . *Left*: dead-seed counts (out of 9 real-baseline runs) per cell; the dashed rectangles mark the matched-shared-trial diagonals. *Right*: maximum step-1 head.weight gradient (log color scale). The dead-seed pattern follows the learning-rate column, with activation modulating sensitivity at high learning rate.

Table 9: Learning-rate-vs-activation disambiguation on synthetic AWGN data. Dead-seed counts and step-1 head.weight gradient (max across 9 = three real families $\times$ three seeds) for each 2×2 cell. Matched-shared-trial regimes are the diagonal cells (crelu@high learning rate, zrelu@low learning rate); off-diagonal cells swap learning rates between activations. Other hyperparameters are held at each activation’s matched selection.

cell	activation	learning rate	dead/9	step-1 head.weight (max)
matched (high)	crelu	0.0236	4/9	31.4
swap (low on crelu)	crelu	0.0024	0/9	2.3
matched (low)	zrelu	0.0024	0/9	0.3
swap (high on zrelu)	zrelu	0.0236	1/9	6.3

selection rule, used widely in CVNN benchmarking to “hold capacity fixed across families,” can manufacture order-of-magnitude accuracy gaps that vanish under independent per-family selection. On our RadioML 3-class subset under crelu, the same data and the same 16-trial search box give:

A $9.3\times$ collapse in the headline gap is not a small bookkeeping effect. The mechanism in Section 3.8.4 and Section 3.8.6 explains why the two rules diverge so sharply on this task: matched-shared-trial picks the validation-best trial *for the complex family* and forces the real baselines to evaluate at that index. When the complex family’s best trial happens to use a learning rate above the explosion-into-dead-region threshold (~ 0.02 here), the real baselines get pushed through a regime that complex tolerates — via its (Re, Im) parameter coupling on the readout — and they do not. Independent selection lets each family avoid that regime, and the apparent advantage shrinks to a few percentage points.

What we recommend. The two rules answer different questions. Matched-shared-trial answers “under a shared hyperparameter allocation, which family is more robust to the chosen regime?” Independent per-family answers “under separate tuning, which family attains the higher peak performance?” Both are legitimate, neither subsumes the other, and reporting only one risks either overstating the gap (matched only) or erasing a real but small effect (independent only). Concretely, we recommend that CVNN-vs-real comparisons report:

1. the matched-shared-trial result, framed as a robustness stress test rather than a peak-performance estimate;
2. the independent per-family result as the best-achievable point estimate; and

Table 10: Selection-rule contrast on the `crelu` configuration: same data, same search space, same 16 trials. Matched-shared-trial holds the trial index fixed across families; independent selection lets each family pick its own best trial. The order-of-magnitude gap difference is the methodological finding, not the size of the complex advantage.

selection rule	complex	best real	gap (pp)
matched-shared-trial	0.7293	0.4999	+22.94
independent per-family	0.7293	0.7047	+ 2.46

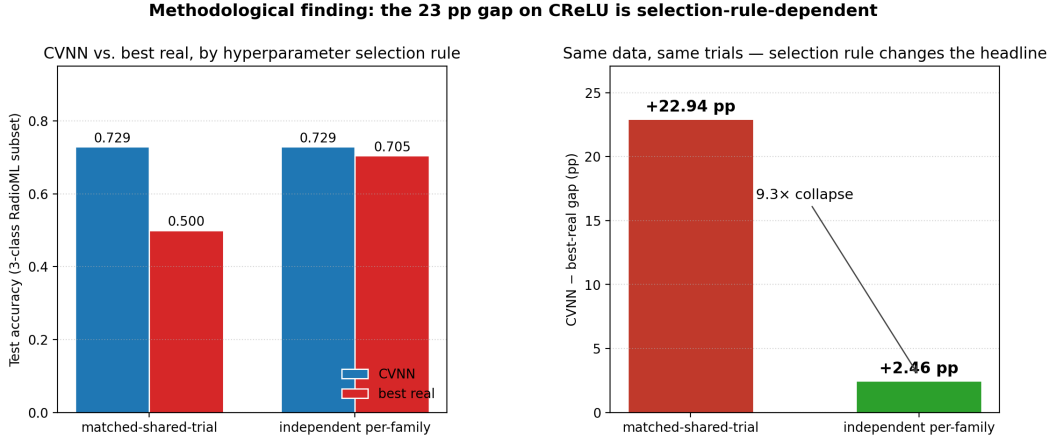


Figure 10: Selection-rule contrast. *Left*: CVNN and best-real accuracies under the two rules. *Right*: the resulting gap collapses by $9.3\times$ when each family is allowed to pick its own best trial. Same data, same 16 trials — the difference is entirely the selection rule.

- the dead-seed count under the matched rule, since the gap between the two rules is bounded below by what the matched rule’s instability mechanism contributes.

A complex-family victory that survives *both* rules is a substantively different claim from one that survives only the matched rule. The CVNN literature, including our own earlier draft, has historically reported only one or the other; the asymmetry between them is itself a finding.

Scope of the artifact. The mechanism we measure is local to readouts that see seed-dependent input scales (here, stacked (Re, Im) features into a ReLU classifier head fed from a Cartesian conv stack) under the matched-trial rule’s tendency to select aggressive learning rates that the complex family tolerates. We do not claim every matched-shared-trial CVNN result in the literature is similarly inflated. We do claim that, in the absence of either an independent per-family rerun or a per-step gradient telemetry trace, a large matched-trial gap is not safely interpretable as a complex-family advantage. Section 3.8.6, Appendix E, and Appendix F give the diagnostic recipe; any future CVNN benchmark that adopts matched-trial selection should run it.

4 Interpretation

The findings together support a more nuanced and falsifiable story than the original “complex wins” framing:

Complex-valued models should be viewed as structured inductive biases for phase- and amplitude-bearing data, not as universally stronger function approximators. They help when the task’s geometry aligns with complex multiplication, they can be matched or beaten by explicit real coordinate systems in other regimes, and they change optimization tolerance under shared hyperparameter stress.

This framing is deliberately application-facing. In RF engineering, the question is rarely whether the samples are “complex” in a formal sense; they are. The practical question is whether the deployment setting needs a model that preserves phase geometry, extracts amplitude geometry, withstands unknown carrier-phase rotations, or trains robustly under limited tuning. The stress tests show that those are separate axes. A PSK classifier, a QAM classifier, and a rotated carrier-offset classifier should not be expected to reward the same architecture. The application pilots make the same point outside RF: wavefunction momentum is invisible to $|\psi|$ but explicit Cartesian/polar coordinates can match native complex layers; EEG analytic signals separate phase locking, amplitude bursts, and phase-amplitude coupling into different winning representations. In both cases, the relevant symmetry or coordinate view has to be tested directly rather than assumed from the complex dtype.

4.1 Why complex helps on IQ data — phase equivariance

The intuition that “complex multiplication has the right semantics for phase-bearing signals” can be made formal in one short lemma.

Lemma 1 (Phase equivariance of `ComplexConv1d`). *Let $C(z)[t] = \sum_{\tau} K[\tau] \cdot z[t - \tau] + b$ be a `ComplexConv1d` layer with complex-valued kernel K , complex bias b , and complex input sequence z . For any global phase $\varphi \in \mathbb{R}$ and any t ,*

$$C(e^{i\varphi} \cdot z)[t] = e^{i\varphi} \cdot C(z)[t] + (1 - e^{i\varphi}) \cdot b.$$

Proof. By linearity of complex multiplication and convolution,

$$C(e^{i\varphi} \cdot z)[t] = \sum_{\tau} K[\tau] \cdot (e^{i\varphi} \cdot z[t - \tau]) + b = e^{i\varphi} \sum_{\tau} K[\tau] \cdot z[t - \tau] + b = e^{i\varphi} \cdot (C(z)[t] - b) + b. \quad \square$$

In the bias-free case (or after subtracting the bias) the result is exactly equivariant. The biased case adds a fixed offset $(1 - e^{i\varphi}) \cdot b$. If the intervening stack preserves the global phase action, the head’s $|\cdot|$ magnitude operation removes it ($|e^{i\varphi} \cdot y| = |y|$ for real φ), yielding phase-invariant logits. Our actual benchmarks include activations that do *not* preserve exact equivariance, so we treat this as an architectural prior supplied by complex multiplication rather than a guarantee that the trained network is globally phase-invariant.

The same does not hold for `Conv1d` operating on a stacked $(\operatorname{Re} z, \operatorname{Im} z)$ real input. A global phase shift $e^{i\varphi}$ corresponds to multiplying each (I, Q) sample by the 2×2 rotation matrix

$$R_{\varphi} = \begin{pmatrix} \cos \varphi & -\sin \varphi \\ \sin \varphi & \cos \varphi \end{pmatrix} = \exp(\varphi J), \quad J = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}.$$

Per-tap equivariance therefore requires every real conv kernel $W_k \in \mathbb{R}^{2 \times 2}$ to commute with R_{φ} for all $\varphi \in \mathbb{R}$. The set of such matrices is much smaller than $\mathbb{R}^{2 \times 2}$ — and turns out to be exactly $\mathbb{C}$.

Proposition 3 (Equivariant 2-channel kernels are complex scalars). *Let $W \in \mathbb{R}^{2 \times 2}$. The following are equivalent:*

1. $WR_{\varphi} = R_{\varphi}W$ for all $\varphi \in \mathbb{R}$;
2. $W = aI + bJ$ for some $a, b \in \mathbb{R}$;
3. W corresponds to multiplication by the complex scalar $a + ib$ under the standard $\mathbb{R}$ -algebra embedding $\iota : \mathbb{C} \hookrightarrow M_2(\mathbb{R})$, $\iota(a + ib) = aI + bJ$.

Proof. (2) $\Leftrightarrow$ (3) is the definition of ι . (2) $\Rightarrow$ (1): $aI + bJ$ commutes with every polynomial in J , including $R_{\varphi} = \exp(\varphi J)$. (1) $\Rightarrow$ (2): differentiating $WR_{\varphi} = R_{\varphi}W$ at $\varphi = 0$ gives $WJ = JW$.

Writing $W = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ and expanding both sides forces $d = a$ and $c = -b$, so $W = aI + bJ$.

(Equivalently: $SO(2)$ acts irreducibly on $\mathbb{R}^2$, so by Schur’s lemma the real commutant is the division algebra $\operatorname{End}_{SO(2)}(\mathbb{R}^2) \cong \mathbb{C}$.) $\square$

Proposition 3 sharpens the “complex layers have an inductive bias” intuition into a precise capacity statement. A two-input-channel real linear map has 4 free parameters per kernel tap; a $U(1)$ -equivariant one has only 2, namely the coefficients (a, b) of $aI + bJ$. The complex layer is exactly

the $U(1)$ -equivariant subspace of the corresponding stacked-real layer, not a richer object: complex multiplication *is* the equivariance constraint, written in coordinates that make the constraint trivial to satisfy. The matched-parameter real baseline of Appendix C is therefore strictly more expressive than the complex layer per tap, but it spends half of those parameters on maps that violate the symmetry; if the task respects $U(1)$, those extra parameters are at best wasted and at worst overfit the asymmetry. This is the structural reason, beyond Lemma 1, that complex layers can win at matched parameters on phase-bearing data without offering any function the real network could not in principle represent.

Relation to group-equivariant deep learning. Proposition 3 is the 1-D specialization of a pattern that is now standard in geometric deep learning [6, 5]: a group action (here $U(1) = SO(2)$) on the input space induces a constraint on the linear map’s commutant; equivariance amounts to choosing parameters inside that commutant. In two spatial dimensions, this is the construction Worrall et al. [28] use in their *harmonic networks*, where complex-valued circular harmonics parameterize $SO(2)$ -equivariant 2-D convolutions; their layer is mathematically the 2-D analogue of our `ComplexConv1d` viewed through Prop. 3, and they observe the same generalization advantage on rotation-bearing tasks. Our paper’s contribution is not the equivariance construction itself, which is classical, but the empirical mapping (Section 3.5) of when this inductive bias helps, hurts, or is irrelevant on phase- and amplitude-bearing 1-D signal data, and the methodological finding (Section 3.9) that some of the apparent advantage in the literature is selection-rule-dependent rather than structural.

This is the structural inductive bias the synthetic phase classification result (Section 3.2), the RF stress tests (Section 3.5), and the RadioML result (Section 3.8) probe in complementary regimes. A single complex sample (phase classification) gives no temporal structure for convolutional equivariance to bite on, and the 2-D real network has the same “see real and imag” feature view, so the complex network buys nothing — the null is what Lemma 1 predicts. A length- L IQ sequence does have temporal structure, but Section 3.5 shows that temporal structure alone is not enough to justify a universal claim: PSK rewards phase-aware geometry, QAM rewards amplitude geometry, and fixed carrier-phase rotation requires either augmentation or an explicit invariance mechanism. The observed wins are consistent with the complex prior when these conditions match, without requiring every activation in the stack to preserve exact equivariance.

What the lemma does not explain. Phase equivariance of the *layer* is necessary for the architectural prior, but the activations between conv layers vary in their own equivariance. By direct computation: `modrelu` and `siglog` are phase-equivariant; `crelu`, `cardioid`, and `zrelu` are not. If the robustness asymmetry in Section 3.8 were *driven* by activation phase equivariance, the stable activations should be exactly `modrelu` and `siglog`. They are not: Section 3.8.4 shows `modrelu` and `zrelu` are stable (0/9 dead seeds across nine real-baseline runs each), while `crelu`, `cardioid`, and `siglog` are unstable (3/9 dead). One phase-equivariant activation is on each side of the divide. The disambiguation in Section 3.8.6 sharpens this further: when we swap learning rates between `crelu` and `zrelu`, the dead-seed pattern follows the lr, not the activation. **Phase equivariance of the layer is necessary for the inductive bias, but the robustness asymmetry Section 3.8.4 measures lives one level lower — at the optimization dynamics induced by (Re, Im) parameter coupling on the classifier head under high-learning-rate AdamW steps, not at the equivariance of the activation.** The two effects are partially independent and partially compounding. This also explains why the Section 3.8 result is most defensible at activations like `zrelu` where complex wins by a small margin rather than at `crelu` where the apparent gap is amplified by the optimization-side mechanism.

5 Related work

Complex-valued neural networks. The mathematical foundations are laid out in Hirose’s textbook [11] and the multi-valued neuron tradition of Aizenberg [1]. The modern deep CVNN era anchors on Trabelsi et al. [23] (`ComplexConv2d`, `ComplexBatchNorm`, the `CReLU/ZReLU/ModReLU` activation triple, and a complex weight initialization), Guberman [9] (early complex-valued CNN for classification), and the unitary RNN line of Arjovsky et al. [2], extended by Wisdom et al. [27] to full unitary capacity. Our paper takes Trabelsi et al. as the canonical CVNN baseline (activation set, default `CReLU`, magnitude readout) but reframes the comparison protocol they used: we show that

the matched-shared-trial selection rule under their CReLU default is the regime in which CVNN-vs-real gaps are most fragile, and we use ZReLU and ModReLU as more conservative reference points.

Wirtinger / CR-calculus and complex backpropagation. Brandwood [4] introduced the complex gradient operator in adaptive array signal processing; the formalism was popularized for ML practitioners by Kreutz-Delgado’s tutorial [12]. PyTorch’s complex autograd [19] adopts the conjugate-Wirtinger convention, which we make explicit in Section 1.3 and verify empirically in our test suite (the rotation-equivariant control of Section 3.4). The Liouville trilemma we state (Proposition 1) is folklore, but to our knowledge no prior CVNN paper presents it alongside an empirical CR-residual measurement of each candidate activation as we do in Section 3.1.

Group-equivariant deep learning. The general construction of G -equivariant convolutional networks was introduced by Cohen & Welling [6] for discrete groups and extended to steerable [7] and general $E(2)$ -equivariant [26] CNNs; the broader geometric DL framework is surveyed in Bronstein et al. [5]. The closest 2-D analogue of our Proposition 3 is the harmonic network of Worrall et al. [28], where complex-valued circular harmonics parameterize $SO(2)$ -equivariant 2-D convolutions. Our Prop. 3 is the 1-D $U(1)$ specialization of that pattern: it identifies the 1-D ComplexConv1d as the 2-parameter equivariant subspace of the unconstrained 4-parameter stacked-real Conv1d, which makes the Trabelsi-style complex layer mathematically continuous with this lineage rather than an unrelated construction.

Complex-valued networks for radio modulation classification. O’Shea et al. [18] introduced the RadioML 2018.01A archive with a real-valued ResNet baseline reaching $\sim 90\%$ top-1 at high SNR on the full 24-class task. Krzyston et al. [13] and Tu et al. [24] report complex-valued AMC architectures with a few-percentage-point edge over real ResNets on the same setup. Our 3-class subset result is consistent with this small advantage *when activations are stable and selection is independent per family*; the larger matched-shared-trial gap we observe under CReLU is a selection-protocol artifact (Section 3.9) rather than a contradiction of the literature, and the per-step mechanism analysis in Section 3.8.4 explains why such artifacts can appear at all.

Complex-valued networks in physics and biomedical signals. CVNNs have been applied to MRI reconstruction with reported gains over real architectures (Cole et al. [8]; Virtue et al. [25]). Physics-informed neural networks for PDE solving (Raissi et al. [20]) typically use real-valued architectures, but the underlying physical fields can be complex; our quantum-wavefunction pilot (Section 3.6) is a small probe in that direction. EEG deep learning has converged on real-valued architectures operating on raw or magnitude-pooled signals (Schirrneister et al. [21], EEGNet [14]); the analytic-signal representation we use in Section 3.7 is standard in the signal-processing literature on phase locking and phase-amplitude coupling [22], but native complex-valued network architectures remain uncommon in EEG. Our pilot is positioned as a representation-protocol demonstration rather than a clinical claim.

Benchmarking methodology and the variance-in-evaluation literature. The selection-rule artifact we document (Section 3.9) belongs to a now-familiar family of findings in which apparent architectural advantages partly dissolve under tighter evaluation protocols: Henderson et al. [10] on deep RL, Lucic et al. [16] on GANs, Melis et al. [17] on language models. Bouthillier et al. [3] formalize the seed-vs.-hyperparameter variance decomposition and recommend reporting both. Our contribution to this literature is specific: we identify the matched-shared-trial selection rule, used widely in CVNN papers as a “capacity-fairness” device, as the operational mechanism, and we provide per-step gradient telemetry (Section 3.8.4, Appendix E) that locates the underlying instability at the readout’s first AdamW step.

6 Limitations

- **RadioML subset, not the full archive.** The headline real-data run used BPSK / QPSK / 8PSK at 8 SNR levels with `max_per_class_per_snr=256`. The full archive has 24 modulations $\times$ 26 SNRs and a much larger per-bucket count.
- **Synthetic QAM is measured, RadioML QAM is not.** The stress tests include QAM16 / QAM64 and show that magnitude can dominate there, but the real-data RadioML mech-

anism study remains PSK-only. A full RadioML QAM/mixed-modulation run is required before making application claims about amplitude-heavy real RF classes.

- **No learned rotation invariance guarantee.** The fixed rotation stress test shows that neither complex layers nor phase-aware real features automatically solve unseen carrier-phase offsets under the current architecture. Rotation augmentation helps but does not make the problem disappear.
- **Application pilots are deliberately small.** The quantum wavefunction study uses 1-D synthetic states, a short split-step evolution, five hand-designed potential families, and three seeds. The EEG study uses synthetic analytic signals rather than clinical sensor data. These are evidence that the representation/stress-test protocol transfers outside RF, not evidence about production scientific or biomedical ML workloads.
- **Seed counts and inference.** The headline RadioML and synthetic RF sweeps use $n = 6$ seeds; the representation stress tests, quantum and EEG pilots, and synthetic phase-classification sweep use $n = 3$. Per-family 95% CIs in $n = 6$ tables now use the t -distribution ($t_{0.975,5} = 2.571$) rather than the Gaussian $1.96\sigma/\sqrt{n}$ approximation; the $n = 3$ pilot tables report means and per-condition stress-test outcomes without CIs since neither Gaussian nor bootstrap CIs are informative at that sample size, and we treat those tables as representation maps rather than significance tests. Where seeds are matched across families (matched-shared-trial selection), we use paired- t tests on per-seed differences, which is both more powerful and more design-appropriate than the unpaired Welch approximation we used in an earlier draft. A bootstrap CI on six seeds gives widths similar to the t -CI and we do not report it separately.
- **One loss.** Classification runs use cross-entropy.
- **No ComplexBatchNorm or ComplexLayerNorm.** Landing them (in the form of Trabelsi et al. [23]’s whitening covariance) would let us train deeper conv stacks fairly.
- **Equivariance characterized at the layer level only.** Proposition 3 and the rotation-equivariant control of Section 3.4 establish equivariance per linear layer. Total network equivariance additionally requires equivariant activations and bias-free or equivariant biases (Appendix A); the fixed-rotation stress test (Section 3.5) and the quantum global-phase shift result (Table 3) document the resulting gap empirically. A construction that is equivariant end-to-end — e.g. stacks of ComplexConv1d with ModReLU or SigLog and bias-free linear readouts — is a natural next step.
- **Modest hardware.** CPU and Apple Silicon for development; one A100 for the headline RF sweep. No multi-machine reproducibility check.
- **Learning-rate-vs-activation disambiguation is 2×2 .** Section 3.8.6 establishes the learning-rate-driven character of the dead-seed threshold using `crelu` (the largest gap) and `zrelu` (the smallest), but does not run the full factorial across all five activations. The cardioid and siglog rows in Table 7 are unaccounted for under this framing except by interpolation; a five-activation learning-rate sweep would close that gap.

7 Future work

- **Scale up the RadioML run** to the full 24-modulation $\times$ 26-SNR archive on the full 1024-sample length. The infrastructure is in place (`sweep_radioml.py -preset full`); the run is deferred from the headline write-up to keep the paper anchored on the 3-class subset that supports the mechanism analysis. Recommended first activation is `zrelu` — the only ablation point where real baselines also train cleanly, so the comparison is least confounded by the matched-shared-trial selection’s interaction with the explosion mechanism.
- **Port the representation stress suite to RadioML.** The synthetic stress tests are already the right scientific shape: PSK-only, QAM-only, mixed modulations, low/high SNR, amplitude normalization, carrier-phase rotation, augmentation, and activation ablation. The next defensible scale-up is to run the same grid on RadioML rather than only increasing class count.
- **ComplexBatchNorm** (whitening 2×2 covariance, per Trabelsi et al. [23]) and deeper conv stacks.

- **Conv2d and 2-D complex domains.** Audio STFT, SAR, MRI [8, 25], and spectrogram-like RF features are natural targets, but they require ComplexConv2d, normalization, and representation stress tests specific to 2-D phase/amplitude geometry. The 2-D analogue of Proposition 3 is the harmonic-network construction [28], which gives a principled starting point.
- **Scale the physics pilot** from 1-D wavefunctions to richer PDE settings: 2-D wave packets, inverse scattering, Fourier-domain operators, and explicit gauge/global-phase invariant architectures.
- **Run the EEG pilot on real analytic-signal data:** sleep staging, seizure detection, brain-computer-interface motor imagery, or anesthesia depth are natural tasks where phase locking, amplitude envelopes, and phase-amplitude coupling [22] have existing signal-processing meaning. The natural baselines are EEGNet [14] and Schirrneister et al. [21], both of which operate on real-valued representations and would let us measure whether the analytic-signal CVNN view adds anything beyond well-engineered real architectures.
- A separate study of per-activation training dynamics: does ModReLU’s bias-sign regime visibly affect convergence?

8 Reproducibility

A C&F submission’s value is in what others can build on. Every claim above is backed by a single CLI invocation against committed artifacts:

```
# 1. Reproduce the benchmark and pilot results
uv sync --all-groups
uv run python experiments/synthetic/sweep_phase_classification.py
uv run python experiments/rf/sweep_synthetic_modulation.py --device cuda
uv run python experiments/rf/representation_stress_tests.py \
    --preset standard --device cuda --resume
uv run python experiments/physics/quantum_wavefunction.py \
    --preset standard --resume
uv run python experiments/neuro/eeg_analytic_signal.py \
    --preset standard --resume
uv run python experiments/rf/sweep_radioml.py --device cuda \
    --activation crelu --seeds 0 1 2 3 4 5
# (repeat for cardioid / siglog / modrelu / zrelu)

# 2. Reproduce the mechanism analysis
uv run python scripts/run_gradient_telemetry.py
uv run python scripts/run_synthetic_rf_telemetry.py

# 3. Regenerate every figure from committed JSON
uv run python scripts/generate_paper_figures.py

# 4. Optional: scale up to the full RadioML archive
uv run python experiments/rf/sweep_radioml.py --device cuda \
    --preset full --activation zrelu --seeds 0 1 2
```

Each result manifest records its `git_commit`, environment, and a `git_dirty` flag computed on a tree that excludes `results/` and the activation-characterization output directory (so regen does not flip the bit on itself). CI surfaces newly added or modified manifests with `git_dirty: true` as warnings on the PR. A structural CI guard checks the committed activation characterization artifact contains all six expected activation rows after a regen.

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A Phase equivariance: full proof and caveats

Lemma 1 states that a bias-free `ComplexConv1d` layer commutes with global phase rotation, and that the bias-bearing case picks up a controlled correction term. We expand the proof and discuss what does and does not propagate through the activation stack.

Proposition 4 (Phase equivariance, bias-free). *Let $C : \mathbb{C}^T \rightarrow \mathbb{C}^T$ denote a one-dimensional complex convolution with kernel $w \in \mathbb{C}^K$, no bias, and zero padding (`padding_mode='zeros'`). For any $z \in \mathbb{C}^T$ and any global phase $\phi \in \mathbb{R}$:*

$$C(e^{i\phi}z)[t] = e^{i\phi} C(z)[t] \quad \forall t \in \{0, \dots, T-1\}.$$

Proof. By linearity over $\mathbb{C}$:

$$C(e^{i\phi}z)[t] = \sum_{k=0}^{K-1} w_k \cdot (e^{i\phi}z)[t-k] = e^{i\phi} \sum_{k=0}^{K-1} w_k \cdot z[t-k] = e^{i\phi} C(z)[t].$$

The phase factor commutes through the sum because complex scalar multiplication is commutative; the sum extracts it cleanly. Boundary indices $t-k < 0$ contribute zero on both sides because zero padding is itself phase-invariant ($e^{i\phi} \cdot 0 = 0$). $\square$

Proposition 5 (Phase equivariance, bias-bearing). *With learnable bias $b \in \mathbb{C}$:*

$$C(e^{i\phi}z)[t] = e^{i\phi} C(z)[t] + (1 - e^{i\phi}) b.$$

Proof. $C(e^{i\phi}z)[t] = e^{i\phi}(C(z)[t] - b) + b = e^{i\phi}C(z)[t] + b - e^{i\phi}b.$ $\square$

What survives the activation stack. Phase equivariance is preserved layer-by-layer if and only if the activation σ itself satisfies $\sigma(e^{i\phi}z) = e^{i\phi}\sigma(z)$.

- **ModReLU** ($\sigma(z) = \text{ReLU}(|z| + b) \cdot e^{i\angle z}$): equivariant. The magnitude term depends only on $|z|$, which is rotation-invariant; the phase factor passes through unchanged.
- **ComplexCardioid** ($\sigma(z) = \frac{1}{2}(1 + \cos \angle z) \cdot z$): *not* equivariant in general, because the gate changes under rotation: $\cos(\angle z + \phi) \neq \cos(\angle z)$ for generic ϕ .
- **Siglog** ($\sigma(z) = z/(1 + |z|)$): equivariant.
- **ComplexTanh** ($\sigma(z) = \tanh z$, the meromorphic extension): *not* equivariant. Direct check: for $z = 1$ and $\phi = \pi/2$, $\tanh(iz) = \tanh(i) = i \tan 1 \approx 1.557i$ while $e^{i\phi} \tanh(z) = i \tanh(1) \approx 0.762i$.
- **CReLU** ($\sigma(z) = \text{ReLU}(\text{Re } z) + i \text{ReLU}(\text{Im } z)$): *not* equivariant. A rotation $e^{i\phi}$ mixes real and imaginary components before the split, elementwise nonlinearity.
- **ZReLU** (zero out unless $\angle z \in [0, \pi/2]$): not equivariant (the kept-quadrant condition is rotation-sensitive).

So four of our six characterized activations break equivariance. The empirical robustness asymmetry we report (a step-1 explosion-into-dead-region pattern that disproportionately hurts real baselines) does *not* track which activations preserve layer-level phase equivariance. CReLU does not preserve equivariance but is the configuration with the largest measured CVNN-vs-real gap; Siglog does preserve equivariance but appears in the unstable high-learning-rate regime. We therefore locate the asymmetry one level lower than the activation, in the (Re, Im) parameter coupling that the optimizer sees on `head.weight` at step 1; Section 3.8.4 develops this in the body.

B Hyperparameter search spaces

Each experiment runs a random search over the box defined below, with N trials and 3 seeds per trial. Selection across baseline families is the matched-shared-trial rule of Section 2: trial i trains all four families on the same hyperparameters, validation selects the trial index, and the test number reports that index across families. The independent-winner table is in Appendix D for diagnostic purposes.

Experiment	Search space	Budget
Synthetic phase classification	hidden $\in \{16, 32, 64\}$, batch $\in \{64, 128\}$, lr $\in [10^{-4}, 10^{-2}]$ log-uniform, steps $\in \{200, 400, 800\}$	16 trials 3 seeds each
Synthetic RF modulation	hidden $\in \{16, 32, 64, 128\}$ (conv) / $\{32, 64, 128\}$ (mlp), batch $\in \{128, 256, 512\}$, lr $\in [10^{-4}, 10^{-2}]$, steps $\in \{200, 400, 800\}$	16 trials 3 seeds
RadioML subset (-preset subset)	hidden $\in \{16, 32, 64\}$ (conv) / $\{32, 64, 128\}$ (mlp), batch $\in \{128, 256, 512\}$, lr $\in [10^{-3}, 5 \times 10^{-2}]$, steps $\in \{200, 400, 800\}$	16 trials 3 seeds
RadioML full (-preset full)	hidden $\in \{32, 64, 128, 256\}$, batch $\in \{256, 512, 1024\}$, lr $\in [10^{-3}, 5 \times 10^{-2}]$, steps $\in \{400, 800, 1600\}$	16 trials 3 seeds (deferred)

Table 11: Search spaces and budgets per experiment. Optimizer is AdamW [15] with weight decay 10^{-4} throughout; learning-rate scheduler is cosine to zero across the trial’s step budget. The full-archive RadioML budget is the deferred scale-up, parameterized via -preset full; everything else is reproducible on a single A100 in under an hour per activation.

Why matched-shared-trial. An independent winner per family selects the family-best hyperparameter from a 16-trial search, which biases up the best of four 16-trial searches and conflates capacity allocation across families. The shared-trial rule fixes the trial index across families, so each family is evaluated at the same allocation; capacity matching is then transparent. We report independent-selection numbers as a diagnostic (Appendix D); the gap between the two selection rules is itself one of the artifacts the C&F framing exposes.

C Architecture details and parameter counts

The four baseline families differ in capacity-allocation rule against the complex network:

- **complex:** `ComplexConv1d(2c→h)` or `ComplexLinear(d→h)`, complex weights $[h \times 2c, K]$ or $[h \times d]$ with 2 floats/parameter (real and imaginary).
- **real_stacked:** `Conv1d(2→h)` or `Linear(2d→h)`; treats (I, Q) as 2 input channels. Same h as the complex network, half the parameter count (1 float per weight).
- **real_matched_params:** hidden width chosen to match parameter count to the complex family. For a complex $(2c \rightarrow h, K)$ layer with $2chK$ complex weights = $4chK$ floats, the matched real layer is $(2 \rightarrow 2h, K) = 4hK$ floats, identical at $c = 1$ but scaling to $\sqrt{2}h$ at $c > 1$.
- **real_matched_flops:** hidden width chosen to match forward-pass multiply-accumulate count rather than parameter count. For our 1-D conv layers the resulting widths track `matched_params` closely (within 5%); listed separately to avoid silently conflating “capacity” senses.

The synthetic phase classifier uses a 1-layer MLP head (no conv); the RF/RadioML classifiers use `ComplexConv1d` (or `Conv1d` on (I, Q) stacked channels) followed by global magnitude pool + linear readout. Magnitude pool is $|z|$ per channel for the complex family and identity for real, since the real channels are already real-valued at the readout.

D Per-activation results, all four baseline families

For each of the five activations used in the RadioML ablation, we report mean $\pm$ standard deviation test accuracy on the -preset subset task at the matched shared-trial winner. The last column reports dead real-baseline seeds across the three real families $\times$ three telemetry seeds, where “dead” means final loss $\geq \ln 3 - 0.05$ (the 3-class chance level). `ComplexTanh` was included in activation characterization but not in the RadioML ablation.

Activation	complex	real_stacked	matched_params	matched_flops	dead real seeds
crelu	0.729 ± 0.009	0.458 ± 0.139	0.500 ± 0.133	0.425 ± 0.142	3/9
cardioid	0.728 ± 0.019	0.457 ± 0.138	0.503 ± 0.135	0.416 ± 0.131	3/9
siglog	0.701 ± 0.016	0.469 ± 0.149	0.466 ± 0.145	0.412 ± 0.123	3/9
modrelu	0.668 ± 0.031	0.632 ± 0.035	0.694 ± 0.062	0.680 ± 0.038	0/9
zrelu	0.733 ± 0.015	0.664 ± 0.030	0.704 ± 0.008	0.699 ± 0.009	0/9

Table 12: RadioML subset, matched-shared-trial selection. The CVNN-vs-real gap is large under three activations (CReLU, ComplexCardioid, Siglog) and small or absent under two others (ModReLU, ZReLU). The gap-bearing activations are exactly those for which the matched-shared-trial hyperparameters produce dead real-baseline seeds; the apparent gap is largely the dead-seed average. See Appendix E for the per-step mechanism.

Independent-winner contrast. The body’s Section 3.9 (Table 10) reports the headline matched-vs-independent contrast on crelu (+22.94 pp → +2.46 pp). The reduction comes from independent selection allowing real baselines to avoid the high-learning-rate trial that triggers the step-1 explosion documented in Section 3.8.4. We do not re-tune the cardioid / siglog ablation rows independently, because the modrelu / zrelu rows already show what stable-lr real selection produces under matched-shared-trial (± 3 pp gaps), and that bracket is the relevant comparison for “what would real baselines do at their preferred lr.”

E Gradient telemetry: per-layer, per-step

Telemetry instruments every parameter tensor with per-step $\|\nabla_{\theta} L\|_F$, the Cauchy-Riemann residual on complex parameters (zero by construction for the complex family), and an intrinsic Jacobian-norm proxy. Runs cover 5 activations $\times$ 4 families $\times$ 3 seeds = 60 instrumented trajectories, each capped at 200 steps for tractability on commodity hardware.

Step-1 head-gradient ratio. Across all five activations, the step-1 ratio $\|\nabla_{\text{head.weight}} L\|_{\text{real}} / \|\nabla_{\text{head.weight}} L\|_{\text{complex}}$ falls into two regimes:

- **Gap-bearing** (CReLU, ComplexCardioid, Siglog): ratio in the range 50–80 at step 1; collapses to ~ 1 by step 20 for surviving seeds, but seeds whose post-step-1 head landed in a dead-ReLU region never produce gradient signal again.
- **Stable** (ModReLU, ZReLU): ratio in the range 1.5–3 at step 1; no seeds enter dead-loss states.

The grouping is consistent across seeds and across the synthetic-data replication (Appendix F). The 50–80 $\times$ figure is a step-1 measurement on raw weights with fan-in initialization; later steps post-Adam-warmup show smaller ratios under all activations. The point of measuring at step 1 is that that is where the dead seeds become dead; past step 1, the parameter state has already diverged.

Why head.weight, specifically. The conv stack’s parameters show no comparable real-vs-complex ratio at any step. The explosion is a property of how matched-trial hyperparameter selection maps into the readout when the readout sees magnitude-pooled inputs (complex) versus stacked (Re, Im) inputs (real). The complex magnitude pool produces inputs of comparable scale across seeds; the stacked (Re, Im) readout sees inputs whose scale is seed-dependent through the conv-stack random init, and high-learning-rate trials interact badly with the unlucky tail of that distribution.

F Synthetic-data replication of the threshold mechanism

To rule out RadioML-specific effects (CFO, multipath, pulse shaping) as the cause of the explosion, we re-run the gradient telemetry harness on synthetic AWGN-only IQ data with the same 4-family / 5-act sweep and the same matched-trial hyperparameters. The synthetic data generator is `experiments/rf/synthetic_modulation.py`, which produces raw-symbol (I, Q) sequences with per-class label imbalance matched to RadioML.

Result. The dead-seed pattern replicates: CReLU, ComplexCardioid, and Siglog produce 2–3 dead seeds per 9 in the real families; ModReLU and ZReLU produce zero. The step-1 head-gradient ratio sits in the same $50\text{--}80\times$ band for the gap-bearing activations. We conclude the mechanism is hyperparameter-driven, mediated by the readout’s input distribution under matched shared-trial selection, and not an artifact of RadioML’s distortions.

G RadioML 2018.01A loader: aliases and strict-SNR mode

Class-label aliasing. The original `classes.txt` distributed with RadioML 2018.01A lists the 24 modulations in an order that does not match the one-hot ordering of the `Y` array; this is a known issue in the community. `experiments/rf/radioml.py` normalizes class labels through a canonical map and prefers `classes-fixed.json` or `classes-fixed.txt` sidecars next to the HDF5 file when present. Common aliases accepted: PSK8 $\rightarrow$ 8PSK, QAM16 $\rightarrow$ 16QAM, APSK16 $\rightarrow$ 16APSK, ASK4 $\rightarrow$ 4ASK, AM labels with underscores (AM_SSB_WC $\rightarrow$ AM-SSB-WC), etc.

Strict-SNR mode. Earlier loader versions silently dropped samples at SNR levels not in the user’s filter list. The default is now `strict_snr=True`, which raises rather than dropping; the test suite includes a regression case that asserts on the strict behaviour. The previous silent-drop behaviour is recoverable via `strict_snr=False` for compatibility with downstream code that intentionally subsets.

H Reproducibility: manifest schema and CI guards

Every long-running script writes a JSON manifest into the run’s output directory (under `results/`). Manifest fields:

- `git_commit`: the SHA at run start.
- `git_dirty`: boolean, computed on a tree filtered to exclude `results/` and the activation-characterization output directory (otherwise regenerating figures from a clean checkout would flip the bit on its own outputs).
- `python`, `torch`, `numpy` versions.
- `cuda`: device string and CUDA version, or "cpu".
- `argv`: full command line.
- `seed`, `n_trials`, `search_space`: the hyperparameter regime.
- `wall_clock_seconds`: end-to-end runtime.

CI surfaces newly-added or modified manifests with `git_dirty: true` as warnings on the PR (initially a hard fail; loosened to a warning at the request of practical iteration). A structural CI guard checks the committed activation characterization artifact contains all six expected activation rows after a regen, so that figure regeneration cannot silently drop an activation.

Caching. Capped sweep subsets cache the seed/filter split to disk and reuse it across trial/family/seed runs. Uncapped full-archive runs disable caching automatically (the working set exceeds typical Mac SSD scratch space); pass `-no-cache-data` to force-disable for capped runs that nevertheless overflow memory.

I Full-archive RadioML scale-up: cost and deferred plan

The submitted experiments are at `-preset subset`: 3 PSK modulations, 8 even SNR levels, `max_per_class_per_snr=256`, `sample_length=128`. The infrastructure for full-archive runs is in place via `-preset full` (24 modulations, 26 SNR levels, `sample_length=1024`); we have deliberately deferred running it because the mechanism analysis is the contribution and the scale-up risks scope creep relative to a C&F submission.

Cost estimate. A single -preset full -activation crelu -seeds 0 1 2 run on an A100 40GB takes 5–15 hours depending on which trial samples land at the upper end of the search space (the search box’s largest configuration is the bottleneck). Six activations $\times$ 3 seeds $\times$ 16 trials $\times$ 4 families = 1152 instrumented runs at full scale; in practice we would run 2–3 activations (CReLU as the headline, ZReLU as the stable counter-example, and one of {ComplexCardioid, Siglog} as a disambiguation). On $4 \times$ A100 in parallel, ~ 24 hours wall-clock for a 3-activation full sweep is a realistic estimate.

What to expect at scale. The C&F prediction is that the two-effect decomposition holds: the architectural prior (phase equivariance, magnitude readout) survives at 24-class / 1024-sample scale, and the optimization-level robustness asymmetry remains hyperparameter-driven and disappears under independent per-family selection. We do not predict the specific numerical gap because the full-archive distribution is qualitatively different from the subset (more distortion-heavy modulations like AM-SSB-WC and FM are present at low SNR), and the matched-trial selection rule may admit different hyperparameter regimes once the conv stack has more channels.

Failure modes to watch for. (1) Caching off may surface stragglers if the loader’s `strict_snr=True` default flags HDF5 SNR levels not in the user’s filter; the loader’s regression test covers this on the subset but the full archive has not been run in CI. (2) A 24-class softmax has higher entropy at chance ($\ln 24 \approx 3.18$ vs $\ln 3 \approx 1.10$), so the dead-loss threshold for detecting failed seeds shifts; the telemetry harness already infers this from the dataset’s class count.

J Activation function catalogue

Activation	Definition	Phase-equivariant?
CReLU	$\sigma(z) = \text{ReLU}(\text{Re } z) + i \text{ReLU}(\text{Im } z)$	no
ZReLU	$\sigma(z) = z$ if $\angle z \in [0, \pi/2]$, else 0	no
ModReLU	$\sigma(z) = \text{ReLU}(z + b) \cdot e^{i\angle z}$	yes
ComplexCardioid	$\sigma(z) = \frac{1}{2}(1 + \cos \angle z) \cdot z$	no
Siglog	$\sigma(z) = z/(1 + z)$	yes
ComplexTanh	$\sigma(z) = \tanh z$ (meromorphic extension; poles at $z = i(\pi/2 + k\pi)$)	no

Table 13: The six activations characterized in this paper, and whether they preserve global phase equivariance layer-by-layer (Appendix A). Equivariance does not predict the empirical CVNN-vs-real gap pattern; five of these activations (all except ComplexTanh) are used in the RadioML ablation, and Siglog’s instability despite equivariance is part of the mechanism evidence in Section 3.8.4.